

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

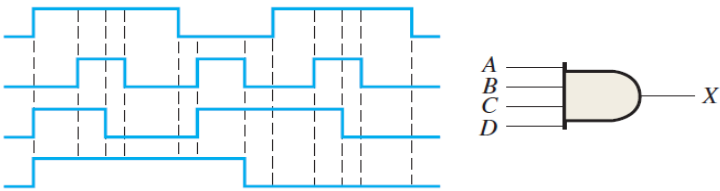
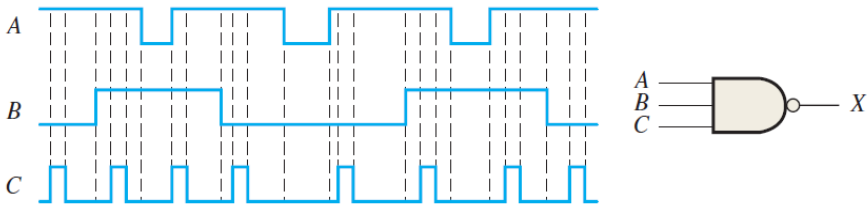
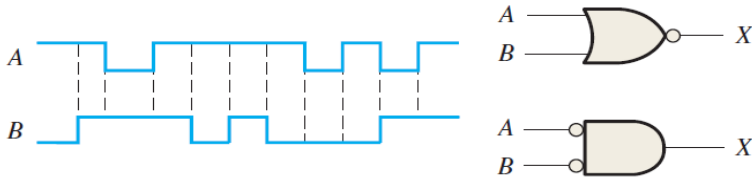
Department of Electrical Engineering

B. Tech. IV Semester

Sub: Electronics Devices & Circuits (EE-223)

Assignment – 3

Submission deadline: 17.04.2025

Q.1.	1) Multiply 01101010 by 11110001 in the 2's complement form. 2) Divide 10001000 by 00100010 in the 2's complement form. 3) Perform $A4_{16} + F5_{16}$ 4) Perform $60_{16} - 39_{16}$
Q.2.	a) Convert the following decimal number to BCD: 1051 b) Convert the BCD numbers to decimal: 0110011001100111 c) Add the following BCD number: $10011000 + 10010111$ d) Convert pair of decimal numbers to BCD, and add as indicated : $295 + 157$
Q.3.	The input waveforms applied to a 4-input AND gate are as indicated in Figure 1. The output of the AND gate is fed to an inverter. Draw the net output waveform of this system.  Fig. 1
Q.4.	Determine the gate output for the input waveforms in Figure 2 and draw the timing diagram.  Fig. 2
Q.5.	The NAND and the negative-OR symbols represent equivalent operations, but they are functionally different. Using functional points of view, show that both gates in Figure 3 produce the same output for the given inputs.  Fig. 3

Q.6.	For the waveforms given in Figure 4, A and B are ANDed with output F; D and E are ANDed with output G ; and C, F, and G are ORed. Draw the net output waveform. Fig. 4
Q.7.	Find the value of X for all possible values of the variables. (a) $X = A + B + C$ (b) $X = (A + B)C$ (c) $X = (A + B)(\overline{B + C})$ (d) $X = (A + B) + (\overline{AB + BC})$ (e) $X = (\overline{A + B})(A + B)$
Q.8.	Apply DeMorgan's theorems to the following: (a) $\overline{\overline{ABC}(\overline{EFG}) + \overline{HIJ}(KLM)}$ (b) $\overline{(A + \overline{BC} + CD) + \overline{BC}}$ (c) $\overline{(\overline{A + B})(\overline{C + D})(\overline{E + F})(\overline{G + H})}$
Q.9.	(a) Draw a logic circuit for the case where the output, ENABLE, is HIGH only if the inputs, ASSERT and READY, are both LOW. (b) Draw a logic circuit for the case where the output, HOLD, is HIGH only if the input, LOAD, is LOW and the input, READY, is HIGH.
Q.10.	Develop the truth table for each of the circuits in Figure 5 Fig. 5
Q.11.	Determine which of the logic circuits in Figure 6 are equivalent

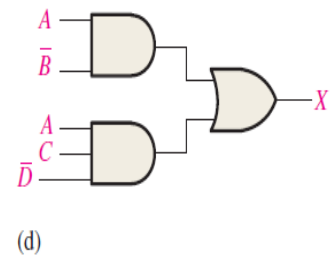
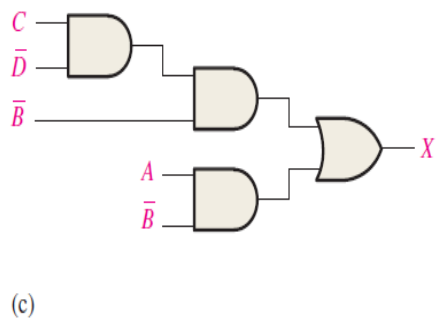


Fig. 6

Q.12.

Convert the following expressions to sum-of-product (SOP) forms:

(a) $(C + D)(A + \bar{D})$ (b) $A(\bar{A}\bar{D} + C)$ (c) $(A + C)(CD + AC)$

Define the domain of each SOP expression and convert the expression to standard SOP form.

Determine the binary value of each term in the standard SOP expressions.

Convert each standard SOP expression to standard POS form.

Q.13.

For each truth table in Table 1 and 2, derive a standard SOP and a standard POS expression.

$ABCD$	X
0000	1
0001	1
0010	0
0011	1
0100	0
0101	1
0110	1
0111	0
1000	0
1001	1
1010	0
1011	0
1100	1
1101	0
1110	0
1111	0

Table 1

$ABCD$	X
0000	0
0001	0
0010	1
0011	0
0100	1
0101	1
0110	0
0111	1
1000	0
1001	0
1010	0
1011	1
1100	1
1101	0
1110	0
1111	1

Table 2

Q.14.	Use a Karnaugh map to reduce each expression to a minimum SOP form: (a) $A + \overline{B\overline{C}} + CD$ (b) $\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + ABCD + ABC\overline{D}$ (c) $\overline{A}B(\overline{C}\overline{D} + \overline{C}D) + AB(\overline{C}\overline{D} + \overline{C}D) + A\overline{B}\overline{C}D$ (d) $(\overline{A}\overline{B} + A\overline{B})(CD + \overline{C}\overline{D})$ (e) $\overline{A}\overline{B} + A\overline{B} + \overline{C}\overline{D} + C\overline{D}$																																																																																										
Q.15.	Use the Karnaugh map method to implement the minimum SOP expression for the logic function specified in truth Table 3. Solve the same for a situation in which the last six binary combinations are not allowed <table><tr><th colspan="4">Inputs</th><th>Output</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> Table 3	Inputs				Output	A	B	C	D	X	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	1	1	0	0	1	0	0	0	0	1	0	1	0	0	1	1	0	1	0	1	1	1	1	1	0	0	0	1	1	0	0	1	0	1	0	1	0	1	1	0	1	1	0	1	1	0	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1
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